

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - PSEUDOCODE WRITING TASK

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	K.KAVIYA	Reg. No.:	192521262
Section / Batch:	1	Date:	28/08/2026

Code of Conduct

I K.KAVIYA(192521262) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.KAVIYA

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

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QUESTIONS

Q1 ✓

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓

Graph - Topological Sort
20 marks

Q3 ✓

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4 ✓

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

```
parents[i] = i;
}
for (int i=0;i<e;i++)
    print(result[i].src,result[i].dest,result[i].weight);
print("total cost:",cost);
}
```

Save Answer

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QUESTIONS

Q1 ✓

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓

Graph - Topological Sort
20 marks

Q3 ✓

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4 ✓

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- Detect cycle before sorting
- Use DFS approach

Your Answer

```
algorithm topologicalsort(G):
    visited = array [v] of 0 // 0:
    stack = []
    function DFS(U):
        visited[u] = 1
        for neighbor v of u:
```

Save Answer

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QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2

Graph - Topological Sort
20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- Use MST for infrastructure
- Use shortest path for routing

Your Answer

```
algorithm smarttransport(G,src):
    T= empty graph
    for each edge (U,V) in G sorted by weight:
        if find (U) != find (V): add(U,V) to T, union (U,V)
    dist[all] = INF , dist[src] = 0 , q = vertices in T
    while q is not empty :
```

Save Answer

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QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2

Graph - Topological Sort
20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
algorithm shortestpathwithvoucher(G,S,D):
    distfromS = dijkstra(G,S)
    distToD = dijkstra(G,D)
    mindistance = distfromS[D]
    for each edge (U,V) with weight W in G:
        cost1 = distfromS[U] + 0 + distToD[V]
```

Save Answer

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)

Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```

for each edge e in MST do:
    G temp = G excudling edge e
    new tree new weight = kruskalorprim(G temp)
    IF new tree is connected AND new weight < min second best THEN:
        min_second_best = new weight
return min_second_best

```

Save Answer